

ISHA KC

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Aerospace Engineering graduate with strong theoretical foundations in aircraft systems, system integration, designing, simulation and airworthiness through academic projects and industry internship irbus Fleet Technical Management.

Education

Pulchowk Engineering Campus, IOE
Bachelor in Aerospace Engineering

2021 – 2025
79.72%

Technical Skills

Systems & Integration: Aircraft systems architecture, subsystem integration, configuration control, maintenance programs, airworthiness concepts

Programming & Tools: MATLAB, C, LaTeX

CFD & Analysis: ANSYS Fluent, OpenFOAM

Aerospace Tools: XFLR5, X-Plane, PlaneMaker, AMASIS

CAD & Modeling: CATIA, OpenVSP

Projects

Preliminary Design and Systems Integration of Point-to-Point Cargo UAV

- Led system-level trade studies integrating payload, propulsion, aerodynamics, and mission requirements.
- Performed mission-based performance analysis, stability assessment, and subsystem sizing.
- Evaluated interfaces between structural layout, aerodynamic configuration, and manufacturability.
- Presented integrated system architecture and performance outcomes through a technical poster.

Low-Speed Wind Tunnel Design and Fabrication

- Designed and fabricated a low-speed wind tunnel as an integrated experimental system.
- Optimized contraction ratio, test section, and flow conditioning for uniform velocity profiles.
- Validated system performance through measurement and comparative analysis.

Oscillating-Vane Gust Generator Development (1-Cosine Profile)

- Designed a mechanically integrated gust generation system to produce controlled 1-cosine longitudinal gusts.
- Worked on actuation mechanism, sinusoidal motion control, and experimental verification.
- Focused on repeatability, system response, and interface between mechanical and aerodynamic components.

Design and Manufacturing of Small-Scale Aircraft

- Designed and integrated aerodynamic, structural, and control aspects of a small-scale aircraft.
- Developed detailed CAD models for CNC-based manufacturing.
- Assessed static stability and configuration feasibility.

1U CubeSat Systems Study and Assembly and structural design

- Participated in system-level study and physical assembly of a 1U CubeSat under ORION Space training with focus on structural design.
- Worked on structural integration of EPS, OBC, and communication subsystems.
- Contributed to interface understanding and packaging constraints.
- Demonstrated the CubeSat model at Mechtrix 2024.

CFD-Based Multiphysics Studies Using OpenFOAM

- Simulated natural convection in a closed cavity to study gravity and thermal boundary effects.
- Modeled stirred tank flow using MRF and $k-\epsilon$ turbulence model.
- Conducted mesh sensitivity, parametric studies, and validation against benchmark data.

Internship Experience

Fleet Technical Management Intern – Airbus India

6 Months (Ongoing)

- Supported Maintenance Program and Configuration Management activities for in-service aircraft.
- Worked on configuration scope definition and part number creation.
- Resolved Airbus FHS queries related to time-controlled components and shelf-life management.
- Reviewed source documents to ensure data consistency across maintenance and configuration systems.
- Gained exposure to airworthiness support, regulatory compliance, and lifecycle data management.

Achievements & Leadership

- Selected for **IIT FOSSEE Summer Fellowship 2025** (OpenFOAM).
- Awarded multiple academic scholarships for academic excellence.
- Recipient of Best Student Award for academic and extracurricular performance.
- Secured 3rd Place in All-Nepal Debate Competition (Physics Initiatives Nepal).
- Scout Troop Leader for 2 years; led a 50-member adventure trail team.
- Literary Coordinator, SEDS Pulchowk.

Additional Activities

- Active member of Initiative for Girls in Physics (IGP), SEDS Pulchowk, and Physics Initiatives Nepal.
- Completed 25+ hours of community service and technical outreach programs.
- Completed a 7-day ANSYS workshop focused on engineering simulation workflows.